

KGOTHATSO NTSANE

Software Engineer — Security, FullStack, Cloud & AI Systems

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PROFESSIONAL SUMMARY

Self-taught developer turned Software Engineer who takes pride in building things that work, with 7+ years shipping production software across media and entertainment, e-commerce, and enterprise — from interfaces to infrastructure, secured by design, for clients including property developers, urban planners, property valuers, the mining industry, architects, and engineers. I threat-model before I code and build systems designed to scale cleanly and stand up to real-world attack. Currently focused on AI-augmented and security-hardened development, type-safe and generic architectures, and shipping faster — and safer — than the next quarterly review. Comfortable owning work end-to-end solo or delivering inside a team.

SELECTED PROJECTS

Insight — AI-Powered Insider Threat Detection (2026) — LIVE: insight.kgothatso.me

Explainable AI for insider threat detection. Stacking ensemble (RF + XGBoost + LightGBM + CatBoost via logistic-regression meta-learner), FGSM/PGD adversarial training, SHAP + LIME dual explainability, MITRE ATT&CK technique mapping with forensic-ready audit trails.

Stack: Python, XGBoost, LightGBM, SHAP, LIME, React

CyberLaw Finder — AI-Powered Cybercrime Law Search (2026) — LIVE: cyberlaw.kgothatso.me

Semantic search for cybercrime statutes across South Africa, USA, and Germany/EU. Sentence-transformers NLP finds relevant laws from plain-language incident descriptions, even without keyword overlap. 96 statutes indexed, 3 jurisdictions.

Stack: Next.js, Python, NLP, sentence-transformers, AI Search

EXPERIENCE

Full Stack Software Engineer — Backend Focus | Nazwo Digital — Remote, Johannesburg (2024 — 2025)

Backend-focused engineering role on a delivery team. Shipped production-grade apps with Python, Docker, and modern JavaScript — going through design reviews, standups, and shared ownership the way real teams work.

- Shipped 5+ production apps
- Design reviews & agile delivery
- System design & API architecture

Stack: Python, JavaScript, React, Node.js, Docker, PostgreSQL

Full-Stack Engineer / DevOps | Open Mic Productions — Southern Africa (2018 — 2024)

Built and maintained full-stack apps serving the media and entertainment industry across Southern Africa. Moved a monolith to microservices on AWS, cut API response times 25% via query optimization + caching, and helped junior engineers find their footing through pairing and reviews.

- Monolith to microservices on AWS
- API latency -25%
- Paired with & mentored juniors

Stack: React, Node.js, MongoDB, AWS, Git, CI/CD

System Admin | SubbieNet (Pty) Ltd. — Pretoria (2015 — 2017)

First IT role. Kept servers and office networks running, helped staff across departments with day-to-day tech issues (-40% resolution time through ticketing), and rebuilt the company site — this is where I learnt that great software = people + code.

- Day-to-day support across departments
- Server & network admin (Windows/Linux)
- Ticketing -40% resolution time

Stack: Windows Server, Linux, Networking, IT Support, Ticketing Systems

SKILLS

Engineering: Frontend Architecture, Generic Programming, Type-Safe Systems, API Engineering, UI & Design Systems, Data Modelling, Performance Optimization, Testing & QA, Microservices, Backend Development, DevOps

Cybersecurity: Network Security, Threat Analysis, Vulnerability Management, Incident Response, Security Frameworks, Risk Assessment, Digital Forensics, Penetration Testing, Security Auditing

AI Systems: Loop Engineering, Agentic Systems, AI Integration, LLM Architecture, Prompt Engineering, Generative AI

Systems: Cloud, Database Design, Infrastructure, Technical Leadership, Container Orchestration, Monitoring & Observability, System Architecture

Stack: React, TypeScript, Node.js, Next.js, Python, PostgreSQL, MongoDB, Docker, Azure, AWS, Git, REST APIs, GraphQL, CI/CD, Tailwind, JavaScript, Express.js, Linux, Redis, Terraform, Figma

Meta Skills: Leading Self, Leading Others, Critical Thinking, Entrepreneurial Thinking, Quantitative Reasoning, Communicating for Impact, Managing Complex Tasks, Navigating Tech Ecosystems

EDUCATION

BScHons in Computer Science (currently completing)	University of Pretoria	2026
Bachelor's Degree in Computing	University of South Africa	2024

CERTIFICATIONS

Google Cybersecurity Professional Certificate	Google	2026
Microsoft Azure Developer Associate (AZ-204)	Microsoft	2026
Microsoft AI Fundamentals (AI-900)	Microsoft	2026
Microsoft Azure Fundamentals (AZ-900)	Microsoft	2025
CompTIA A+ (Mobile Engineering)	CompTIA	2025
Asana Workflow Specialist	Asana	2025
Google Analytics Certification	Google	2024
Full Stack Software Engineering (Back End)	ALX Holberton	2025

References available on request. Portfolio: kgothatso.me